

SymPy Bug Card

Bug 53: singularities Returns EmptySet for zeta(x) at Its Pole

Task. Find the complex singularities of the Riemann zeta function as a function of x .

$$\text{singularities}(\zeta(x), x, \mathbb{C})$$

SymPy's answer.

$$\text{singularities}(\zeta(x), x, \mathbb{C}) = \emptyset.$$

Correct answer.

$$\text{singularities}(\zeta(x), x, \mathbb{C}) = \{1\}.$$

Explanation. The Riemann zeta function is meromorphic with a simple pole at $x = 1$. SymPy itself confirms the pole through $\zeta(1) = \text{zoo}$ and $\lim_{x \rightarrow 1} \zeta(x) = \text{zoo}$, so returning $\emptyset$ omits a genuine singularity.

Likely root cause. The diagnosed implementation of `singularities` scans negative powers and a short list of elementary functions, but it does not consult special-function pole information such as the known zeta pole at $s = 1$. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-053-singularities-returns-emptyset-for-zeta-x-at-its-pole/](#)